

Requirement Analysis Phase

Solution Requirements (Functional & Non-functional)

Date	13 July 2026
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection & Extraction	Connect housing dataset (CSV) to Tableau Verify column data types and row count
FR-2	Data Preparation	Check dataset for null values Check dataset for duplicate records
FR-3	Data Visualization	Build KPI summary sheet (count, average price, basement area) Build sales-by-renovation chart Build house age distribution pie chart Build bedrooms/bathrooms/floors age chart
FR-4	Dashboard Creation	Combine all visualizations into one dashboard Add interactive filters Format layout for readability
FR-5	Storyboard Creation	Build guided story with multiple points Add captions explaining each insight
FR-6	Performance Testing	Document amount of data loaded Document filters used Document number of visualizations
FR-7	Documentation & Demonstration	Record end-to-end explanation video Write step-by-step project documentation

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard should be easy to navigate with clear titles and labels so non-technical viewers can interpret insights without training.
NFR-2	Security	Dataset and workbook are shared only through controlled links (GitHub repo, Tableau Public); no sensitive personal data is exposed.
NFR-3	Reliability	Visualizations consistently display values matching the underlying dataset every time the workbook is opened.
NFR-4	Performance	Dashboard loads and responds to filter changes within a few seconds, given the dataset size (~21,600 rows).
NFR-5	Availability	Published Tableau Public dashboard and GitHub repository remain accessible via their links at all times for mentor review.
NFR-6	Scalability	Dashboard structure can accommodate additional visualizations or a larger dataset in the future without a full redesign.